ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

AUTOMATION AND ROBOTICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 3rd year final examination: PLCs and ladder logic, pneumatic actuators and sequencing, robot fundamentals and coordinate transformations, forward and inverse kinematics of a planar arm, the Jacobian, trajectory planning, mobile robots, encoders and robot safety.

Written by Academic Central for practice, following robotics and automation modules at South African universities (e.g. University of Pretoria ERB 410 Robotics: principles, modelling and control of robotic systems), with PLC and pneumatic automation as taught in mechatronics programmes. It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Draw ladder diagrams neatly, using IEC symbols. Show ALL your working in calculations.
3. Angles are in degrees unless radians are asked for. Give answers correct to THREE decimal places for lengths in metres and TWO decimal places otherwise.
4. Non-programmable scientific calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: PLCs AND LADDER LOGIC

1.1 Describe the scan cycle of a PLC. (3)

1.2 Explain the difference between sinking and sourcing 24 V DC PLC inputs. (2)

1.3 A conveyor motor M is started with a normally open START button and stopped with a normally closed STOP button. A normally closed motor-overload contact OL must also stop it. Draw the ladder rung, with a latching (seal-in) contact. (5)

1.4 Extend the program so that the conveyor stops by itself 10 s after it is started. Describe the extra rung(s) using an on-delay timer (TON). (4)

1.5 A counter (CTU) must count 12 bottles passing a sensor, then stop the conveyor and switch on a "box full" lamp. Describe the rungs needed, including how the counter is reset. (4)

1.6 Name TWO programming languages defined in IEC 61131-3 other than ladder diagram. (2)

[20]

QUESTION 2: PNEUMATIC ACTUATORS

2.1 Explain the difference between a single-acting and a double-acting cylinder. (2)

2.2 A double-acting cylinder has a bore of 50 mm and works at 6 bar. Calculate the theoretical extending force. (3)

2.3 Two cylinders must follow the sequence A+ B+ A− B−. Explain how limit switches and the PLC make sure each step starts only when the previous one has finished. (3)

2.4 What does the description "5/2 single-solenoid, spring-return valve" mean? (2)

[10]

QUESTION 3: ROBOT FUNDAMENTALS AND TRANSFORMATIONS

3.1 Define degrees of freedom and workspace. Name the two basic joint types, and compare a SCARA robot with a six-axis articulated robot. (6)

3.2 Write the rotation matrix for a rotation of 30° about the z-axis, and use it to rotate the point (1; 0; 0). (4)

3.3 Frame B is rotated 90° about the z-axis of frame A, and the origin of B is at (2; 1; 0) in frame A. A point has coordinates (1; 0; 0) in frame B. Use a homogeneous transformation to find its coordinates in frame A. (5)

[15]

QUESTION 4: KINEMATICS OF A PLANAR ARM

FIGURE 1 shows a two-link planar robot arm with L1 = 0.5 m and L2 = 0.3 m. θ1 is measured from the x-axis and θ2 from the line of link 1.

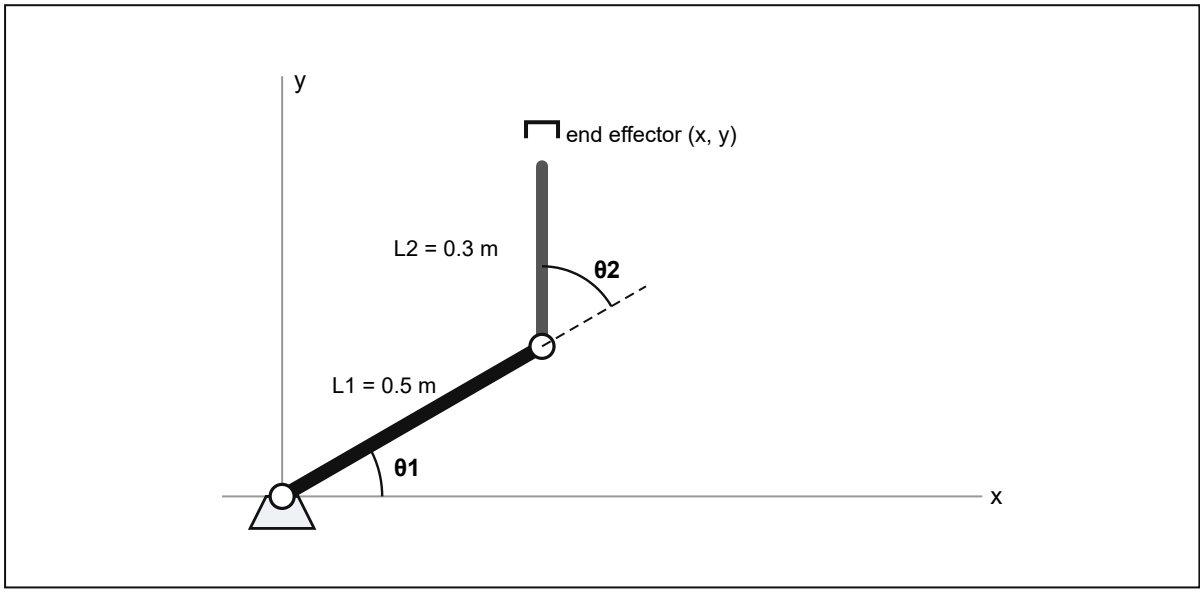


FIGURE 1: Two-link planar arm (drawn at θ1 = 30°, θ2 = 60°)

4.1 Write the forward-kinematics equations for the position (x; y) of the end effector. (3)

4.2 Calculate the position of the end effector when θ1 = 30° and θ2 = 60°. (4)

4.3 Inverse kinematics: the end effector must reach the point (0.6; 0.3) m.

4.3.1 Calculate θ2 for the elbow-down solution (θ2 positive). (4)

4.3.2 Calculate θ1 for this solution. (4)

4.3.3 Give the second (elbow-up) solution. (2)

4.4 Describe the workspace of the arm, and say whether it can reach the point (0.9; 0) m. (3)

4.5 The Jacobian relates joint speeds to end-effector velocity.

4.5.1 Derive the Jacobian matrix of the arm. (3)

4.5.2 Evaluate it at θ1 = 30°, θ2 = 60°. (2)

4.5.3 Calculate the end-effector velocity when $\theta1' = 0.5$ rad/s and $\theta2' = 1$ rad/s. (3)

4.5.4 For which values of θ2 is the arm in a singular position, and what does this mean? (2)

[30]

QUESTION 5: TRAJECTORIES, MOBILE ROBOTS AND SAFETY

5.1 A joint must move from 0° to 90° in 3 s, starting and ending at rest.

5.1.1 Find the cubic polynomial $\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$. (5)

5.1.2 Calculate the largest joint speed. (2)

5.2 Explain independent-joint PD control, and why gravity compensation is added for a robot arm. (4)

5.3 A differential-drive robot has wheels of radius 0.05 m, 0.3 m apart. The right wheel turns at 10 rad/s and the left wheel at 6 rad/s.

5.3.1 Calculate the forward speed and the turning rate of the robot. (4)

5.3.2 Calculate the radius of the circle it drives in. (2)

5.4 A wheel encoder gives 500 pulses per revolution on each of two channels (quadrature). Calculate the counts per revolution with 4× decoding and the angular resolution. (3)

5.5 Name THREE safety measures for an industrial robot cell, and explain what makes a collaborative robot (cobot) different. (5)

[25]

TOTAL: 100

END OF QUESTION PAPER